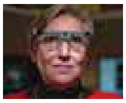
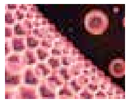


Brain implant to restore vision
Brain implant gives blind woman artificial vision in scientific first.
https://dgit.in/nov21-15



Anti-ageing soon a reality?
Scientists discover new enzymatic complex that can stop cells from aging.
https://dgit.in/nov21-16

LIFE EXTENSION

Are the first immortals already born?

Aditya Madanapalle | aditya@digit.in

The march of technology is extending lifespans, and researchers anticipate an increase in the number of old people around the world over the next few decades. While some old people manage to maintain healthy, functional bodies and minds, others deteriorate in many ways, from being forgetful, to losing motor control of their bodies, to a fading out of the senses. While the body deteriorates as a whole, individual cells also can grow old, accumulating errors in their genetic codes, slowing down tissue growth, and being susceptible to cancer. On a cellular level as well as an individual level, aging is known as senescence in scientific terms. The question researchers are asking is, if senescence can be treated as a disease, and if so, can we cure it?

Hermann Joseph Muller, one of the first to warn of the link between radiation and mutation, studied mutations in drosophila (fruit flies) that had been bombarded with X-rays in the 1930s. Muller noted that the ends of the chromosomes were strangely resistant to the mutations and deterioration, and studied these end segments, coining the word telomere, from the Greek roots of "end" and "part". In the 1940s and 50s, Barbara McClintock meticulously studied mutations in corn, uncovering many fundamental mechanisms of genetics, and showed that the telomeres had an important role to play in the preservation of genetic information. Muller won the 1946 Nobel prize in physiology or medicine, while Barbara McClintock won it in 1983. The 2009 Nobel prize in physiology or medicine was shared by Elizabeth H. Blackburn, Carol W. Greider and Jack W. Szostak, who between them figured out how telomeres work. Blackburn studied the chromosome of a unicellular eukaryote, Tetrahymena, and noticed a DNA sequence that repeated multiple times at the end of chromosomes, CCCCAA. Szostak who was parallelly conducting experiments, noticed that linear DNA molecules, so called "minichromosomes" would rapidly deteriorate when introduced into yeast cells. Blackburn and Szostak combined their researchers, tagged the minichromosomes with the repeated sequences, then introduced them to the yeast cells, and noticed that the telomeres were preventing the DNA molecules from deteriorating. As the DNA from the Tetrahymena was able to fend off the deterioration in a totally different organism, yeast, the